

Unified Microscopic Foundations of the Quantum Boltzmann Equation

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The quantum Boltzmann equation (QBE), originally formulated as the quantum extension of Boltzmann's classical transport theory by Uehling and Uhlenbeck and subsequently developed within the nonequilibrium Green-function formalisms of Kadanoff–Baym and Keldysh, provides the central kinetic framework for describing relaxation and transport in weakly interacting quantum many-body systems. Although its formal structure is well established, conventional derivations often obscure the microscopic equivalence between operator and nonequilibrium Green-function approaches. Wick factorization is typically applied without explicitly enumerating real-time contractions, while Green-function treatments encode the same physics in Born-level self-energies that conceal their operator origin.

In this paper, we present a pedagogical microscopic derivation of the quantum Boltzmann equation that makes this equivalence fully explicit. Starting from the ladder-operator formulation, we enumerate all fifteen Wick pairings of the second-order six-field correlator, identifying three Hartree–Fock contributions and twelve real-time Born scattering channels generated by a quartic, number-conserving interaction Hamiltonian. We then establish a one-to-one correspondence between these twelve operator contractions and the Keldysh components of the second-order Born self-energy, demonstrating explicitly how the Uehling–Uhlenbeck collision integral emerges after Wigner transformation and quasiparticle projection. Under controlled Born–Markov and quasiparticle approximations, this work offers a transparent synthesis of operator and Green-function formulations, and clarifies how kinetic fluctuations can be understood as emergent under coarse-graining.

Keywords: Quantum Boltzmann equation; quantum kinetic theory; nonequilibrium green functions; Keldysh formalism; Wick contractions; born approximation; collision integral.

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1. Introduction

The quantum Boltzmann equation (QBE) provides the central kinetic framework for describing relaxation and transport in quantum many-body systems. By adapting a kinetic equation to the quantum statistics introduced by Nordheim [1], namely, Bose–Einstein and Fermi–Dirac statistics, a Boltzmann-type kinetic equation was initiated for quantum systems. Both quantum statistics are describable in a unified way by introducing statistical occupation factors into phase space, where single-particle motion, external forces and collision-like interactions are incorporated into a single probabilistic evolution equation. The formal structure of the QBE with scattering considerations was later established by Uehling and Uhlenbeck in 1933 [2], who extended the classical Boltzmann collision integral to account for Bose enhancement and Pauli blocking effects. Despite its phenomenological success, the Uehling–Uhlenbeck equation did not provide a microscopic derivation based on underlying quantum field operators.

The operator foundations required for such a derivation were laid by Wick [3], who introduced the contraction algebra and established the factorization theorem for time-ordered products of quantum fields. According to Wick’s theorem, a six-field correlator has 15 distinct pairings, of which three correspond to Hartree–Fock terms, leaving twelve nontrivial contractions. In the second-order Born approximation, involving two interaction vertices and six field operators, these contractions combine with real-time ordering and the gain-loss structure of the kinetic equation to generate twelve distinct real-time contraction channels. These channels encode the microscopic structure of scattering amplitudes, quantum-statistical occupation factors and gain–loss processes, thereby providing the operator-level foundation of quantum kinetic theory.

A complementary perspective is provided by the Green-function formulation originating from Schwinger’s variational principle [4], which yields the equations of motion for time-ordered propagators. Building on this approach, Kadanoff and Baym extended the formalism to interacting nonequilibrium systems and derived exact coupled equations of motion for the one- and two-particle Green functions, G and G_2 [5]. These equations form the modern foundation of nonequilibrium quantum dynamics. In direct analogy with the operator-based treatment, Kadanoff and Baym observed that a kinetic equation emerges upon replacing G_2 by its Wick-factorized form. However, the intermediate algebraic steps linking the contraction structure of G_2 to the corresponding Born self-energy were not made explicit.

A further advance was achieved with Keldysh’s closed-time-path formalism [6], which introduced the real-time matrix structure of Green functions and the lesser and greater components ($<, >$) that encode nonequilibrium distributions and quantum noise. This formulation enables the Dyson equation to be expressed directly in real time and allows collision kernels to be identified as specific combinations of retarded, advanced, lesser and greater propagators. Langreth and Wilkins subsequently developed the projection rules that map contour-ordered products onto their

real-time components [7]. In particular, their identity for triple contour products [7, Eq. (A.12)] provides the essential algebraic tool for projecting the Wick-factorized two-particle Green function G_2 onto the second-order Born self-energies that govern kinetic evolution.

Moyal's phase-space formulation of quantum mechanics [8], based on the Wigner transform, established a direct connection between quantum dynamics and the classical Liouville description through the Moyal bracket. This framework justifies the gradient expansion used to obtain semiclassical kinetic equations from the Wigner transformed Kadanoff–Baym equations.

The historical development of quantum kinetic theory reflects a progressive refinement of microscopic control. Early phenomenological extensions of classical kinetics to quantum systems were introduced by Nordheim and by Uehling and Uhlenbeck, incorporating quantum statistics at the level of scattering processes. Wick subsequently provided the operator algebra necessary for fully microscopic derivations, while Kadanoff and Baym embedded these ideas within a systematic nonequilibrium Green-function hierarchy. The explicit treatment of real-time nonequilibrium dynamics was later achieved through Keldysh's closed-time-path formalism. The quantum Boltzmann equation emerges only after these formalisms are supplemented by controlled approximations. In particular, within the Born truncation, Markov limit and quasiparticle approximation, the ladder-operator and nonequilibrium Green-function approaches represent equivalent realizations of the same underlying microscopic kinetic content.

Despite these developments, most standard presentations of the quantum Boltzmann equation compress the intermediate algebraic structure. In widely used treatments, including those based on the Kadanoff–Baym and Keldysh formalisms as well as standard texts on nonequilibrium and semiconductor transport [9], the Wick factorization of the two-particle Green function G_2 is typically invoked without explicitly enumerating the full set of real-time contractions. Instead, the connection to the Born self-energy is usually expressed in compact diagrammatic form. While the microscopic equivalence between the operator (ladder) formulation and the nonequilibrium Green-function (NEGF) approach is well understood at a conceptual level, it is rarely presented explicitly in terms of individual contractions, scattering channels and their associated momentum-transfer processes.

This review is organized to make the microscopic connections between different formulations of quantum kinetic theory fully explicit. We present a detailed derivation of the quantum Boltzmann equation that (i) enumerates all twelve Wick contractions contributing at second order; (ii) projects each contraction onto real time using the Langreth–Wilkins rules, thereby generating twelve distinct real-time scattering channels; (iii) constructs the direct and exchange contributions to the second-order (Born) correlation self-energy; and (iv) demonstrates explicitly how these channels reconstruct the Uehling–Uhlenbeck collision integral after Wigner transformation and the quasiparticle approximation. Although the equivalence between operator-based and nonequilibrium Green-function formulations is well

established in principle, the real-time component form at the level of individual contractions and scattering channels is almost non-existent in literature.

1.1. *Scope and limitation*

This review is restricted to weakly interacting systems admitting a quasiparticle description, while strongly correlated regimes, integrable systems and dynamics near nonthermal fixed points lie outside the present scope and typically require frameworks beyond the Markovian quantum Boltzmann approach.

The quantum Boltzmann equation derived in this work is valid within a controlled hierarchy of approximations. These include weak coupling, such that binary scattering dominates (Born truncation); a separation of time scales, allowing a Markovian limit; and the existence of well-defined quasiparticles, permitting an on-shell description of occupation dynamics. Within this regime, the collision integral provides a consistent reduction of the underlying many-body dynamics. When these assumptions fail — such as in strong-coupling systems, regimes with long memory, broad spectral functions, bound states, or near-critical dynamics — one must instead retain non-Markovian kernels, off-shell transport, or employ resummation schemes beyond the quantum Boltzmann framework. These extensions lie outside the scope of the present review.

Throughout this work, symbol U denotes a generic effective interaction kernel entering a quartic, number-conserving interaction Hamiltonian. For particle systems, U may represent a physical two-body scattering potential. For field-based systems, including electromagnetic fields, U should instead be understood as the momentum-space representation of an effective four-wave-mixing vertex obtained after eliminating fast microscopic degrees of freedom. In nonlinear optics notation, this interaction is commonly expressed in terms of a coupling coefficient λ_{1234} . These descriptions represent equivalent formulations of the same quartic interaction and differ only by Fourier conventions and the associated momentum-conserving delta functions. No assumption is made that photons scatter from a mechanical potential.

Finally, we emphasize that this review focuses on the microscopic derivation of the quantum Boltzmann equation for systems governed by a quartic, number-conserving interaction Hamiltonian. This class includes dilute quantum gases and weakly nonlinear fields dominated by four-wave-mixing processes. Interactions that do not conserve particle number such as parametric down-conversion, harmonic generation, or multiphoton absorption, require extensions of the Hamiltonian and give rise to additional source and sink terms beyond the gain–loss structure derived here. While the derivation is presented in a general many-body setting, its explicit contraction structure is directly relevant to quantum plasma kinetics, where the collision integral is often assumed rather than derived. Spin degeneracy factors are suppressed for clarity and can be reinstated multiplicatively without altering the structure of the kinetic equation.

2. Ladder-Operator Derivation of the Quantum Boltzmann Equation

The derivation presented in this section is not intended as a new result, but rather as a pedagogical reconstruction designed to make explicit the Wick contraction structure typically compressed in standard treatments.

In this section, we present an operator-based derivation of the quantum Boltzmann equation, starting directly from the Heisenberg equations of motion for ladder operators. The objective is to expose the microscopic scattering structure, namely the underlying matrix elements, conservation laws and gain–loss factors, entirely within the operator formalism. This derivation provides a transparent baseline that will later be reproduced within the nonequilibrium Green-function framework in Sec. 3.

At second order in the interaction, the time evolution of the single-particle occupation number is governed by a six-operator correlator. Applying Wick’s theorem, the six-operator correlator decomposes into fifteen distinct pairings, of which three correspond to Hartree–Fock (mean-field) contributions and the remaining twelve represent genuine scattering processes that generate the full gain–loss structure of the quantum Boltzmann collision integral. The explicit contraction algebra is presented in [Appendix A](#).

2.1. Derivation

Here, we derive the collision integral of the QBE by employing Wick’s theorem, which provides a framework for decomposing expectation values of products of multiple field operators into pairwise contractions, thereby enabling a controlled microscopic treatment of interaction-induced scattering processes. We consider a quartic, number-conserving interaction Hamiltonian describing elastic two-body collisions:

$$\hat{H}_{\text{int}} = \frac{1}{2} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{p}_1 - \mathbf{p}_2) \hat{a}_{\mathbf{p}'_1}^\dagger \hat{a}_{\mathbf{p}'_2}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1}, \quad (1)$$

where the summation is restricted to momentum ($\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{p}'_1 + \mathbf{p}'_2$) and energy conservation ($\frac{p_1^2}{2m} + \frac{p_2^2}{2m} = \frac{p'^2_1}{2m} + \frac{p'^2_2}{2m}$), and the scattering amplitude satisfies the symmetry condition $U(\mathbf{p}) = U(-\mathbf{p})$. No assumption is made that U represents a mechanical potential; it denotes a generic quartic interaction vertex appropriate to the quasiparticle description. In classical kinetic theory, the Boltzmann equation is formulated in terms of a phase-space distribution function $f(t, r, v)$, reflecting a semiclassical picture in which position and momentum are simultaneously well defined. For quantum systems, such a description requires an appropriate averaging procedure to account for the constraints imposed by the uncertainty principle. This leads naturally to a phase-space formulation based on the Wigner representation. We define the semiclassical quantum distribution function $f(t, r, p)$

as the spatial average

$$\begin{aligned} f(t, \mathbf{r}, \mathbf{p}) &= \frac{1}{(2\pi)^3} \int d\mathbf{r}' e^{-i\mathbf{p}\cdot\mathbf{r}'} \left\langle \hat{\Psi}^\dagger \left(t, \mathbf{r} - \frac{1}{2}\mathbf{r}' \right) \hat{\Psi} \left(t, \mathbf{r} + \frac{1}{2}\mathbf{r}' \right) \right\rangle \\ &= \frac{1}{(2\pi)^3} \sum_{\mathbf{k}, \mathbf{k}'} \delta \left(\mathbf{p} - \frac{\mathbf{k} + \mathbf{k}'}{2} \right) e^{-i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \left\langle \hat{a}_{\mathbf{k}}^\dagger(t) \hat{a}_{\mathbf{k}'}(t) \right\rangle, \end{aligned} \quad (2)$$

where $\hat{\Psi}(t, \mathbf{r}) = \sum_{\mathbf{k}} \hat{a}_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{r}} / (2\pi)^{3/2}$ denotes the field operator at spacetime point $(t, \mathbf{r})$. It is pertinent to mention here that for spatially homogeneous systems, the phase-space distribution function $f(t, \mathbf{r}, \mathbf{p})$ reduces up to normalization to the mode occupation number, i.e.,

$$\begin{aligned} n_{\mathbf{p}}(t) &= \int d\mathbf{r} f(t, \mathbf{r}, \mathbf{p}) = \frac{1}{(2\pi)^3} \int d\mathbf{r} \sum_{\mathbf{k}, \mathbf{k}'} \delta \left(\mathbf{p} - \frac{\mathbf{k} + \mathbf{k}'}{2} \right) e^{-i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \left\langle \hat{a}_{\mathbf{k}}^\dagger(t) \hat{a}_{\mathbf{k}'}(t) \right\rangle \\ &= \sum_{\mathbf{k}, \mathbf{k}'} \delta \left(\mathbf{p} - \frac{\mathbf{k} + \mathbf{k}'}{2} \right) \delta(\mathbf{k} - \mathbf{k}') \left\langle \hat{a}_{\mathbf{k}}^\dagger(t) \hat{a}_{\mathbf{k}'}(t) \right\rangle = \sum_{\mathbf{k}} \delta(\mathbf{p} - \mathbf{k}) \left\langle \hat{a}_{\mathbf{k}}^\dagger(t) \hat{a}_{\mathbf{k}}(t) \right\rangle \\ &= \left\langle \hat{a}_{\mathbf{p}}^\dagger(t) \hat{a}_{\mathbf{p}}(t) \right\rangle. \end{aligned} \quad (3)$$

The operator manipulations that follow are standard and are included here solely to make explicit the microscopic origin of the gain–loss structure of the quantum Boltzmann collision integral. In the interaction picture, the Heisenberg equation of motion for the operator product $\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}'}$ appearing in Eq. (2) is given by

$$\begin{aligned} \frac{d}{dt} \left(\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}'} \right) &= \frac{i}{2\hbar} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_1', \mathbf{p}_2'} U(\mathbf{p}_1 - \mathbf{p}_2) \left\{ \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_2 \mathbf{k}} + \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_1 \mathbf{k}} \right. \\ &\quad \left. - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_1' \mathbf{k}'} - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_2' \mathbf{k}'} \right\}. \end{aligned} \quad (4)$$

Here, we employ the commutation relation $\hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{q}}^\dagger = \delta_{\mathbf{p}\mathbf{q}} + \eta \hat{a}_{\mathbf{q}}^\dagger \hat{a}_{\mathbf{p}}$, where $\eta = +1$ for bosons and $\eta = -1$ for fermions. To obtain a closed kinetic equation, Eq. (4) must be evaluated consistently to second order in the interaction Hamiltonian. At this order, the operators on the right-hand side are expanded to first order in $\hat{H}_{int}$ using $\hat{F}(t) \simeq \hat{F}(0) + \frac{i}{\hbar} \int_0^t dt' [\hat{H}_{int}(t'), \hat{F}(0)]$. Up to this point, the dynamics remains time-reversal invariant: neither the Born expansion nor Wick factorization introduces irreversibility.

At second order, the time derivative of the occupation number is governed by a six-field correlator. Application of Wick's theorem decomposes this correlator into 15 distinct pairings, three of which correspond to Hartree–Fock (mean-field) renormalizations, while the remaining twelve represent genuine two-particle scattering processes. These twelve Wick contractions generate the full gain–loss structure of the quantum Boltzmann collision integral. The explicit contraction algebra is collected in [Appendix A](#). Here, we evaluate the sums over $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_1', \mathbf{q}_2'$, using the Kronecker deltas and use the symmetry $U(\mathbf{p}) = U(-\mathbf{p})$. Relabeling dummy indices in the

second energy-conserving term of the last equation of [Appendix A](#) according to $\mathbf{p}_1 \rightarrow \mathbf{p}'_1$, $\mathbf{p}_2 \rightarrow \mathbf{p}'_2$, $\mathbf{p}'_1 \rightarrow \mathbf{p}_1$ and $\mathbf{p}'_2 \rightarrow \mathbf{p}_2$, we obtain

$$\begin{aligned} \frac{d}{dt} \langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}'} \rangle &= \frac{2i}{\hbar} \sum_{\mathbf{p}_1} U(\mathbf{p}_1 - \mathbf{k}) n_{\mathbf{p}_1} (n_{\mathbf{k}'} - n_{\mathbf{k}}) - \frac{4\pi}{\hbar^2} \sum_{\mathbf{p}_1, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{k} - \mathbf{p}_1) U(\mathbf{p}'_2 - \mathbf{p}'_1) \\ &\quad \times \{ \delta(\omega_{\mathbf{k}'} + \omega_{\mathbf{p}_1} - \omega_{\mathbf{p}'_1} - \omega_{\mathbf{p}'_2}) [n_{\mathbf{k}'} n_{\mathbf{p}_1} - n_{\mathbf{p}'_1} n_{\mathbf{p}'_2} + \eta \\ &\quad \times (n_{\mathbf{k}'} n_{\mathbf{p}_1} (n_{\mathbf{p}'_1} + n_{\mathbf{p}'_2}) - n_{\mathbf{p}'_1} n_{\mathbf{p}'_2} (n_{\mathbf{k}'} + n_{\mathbf{p}_1}))] \\ &\quad + \delta(\omega_{\mathbf{k}} + \omega_{\mathbf{p}_1} - \omega_{\mathbf{p}'_1} - \omega_{\mathbf{p}'_2}) [n_{\mathbf{k}} n_{\mathbf{p}_1} - n_{\mathbf{p}'_1} n_{\mathbf{p}'_2} + \eta \\ &\quad \times (n_{\mathbf{k}} n_{\mathbf{p}_1} (n_{\mathbf{p}'_1} + n_{\mathbf{p}'_2}) - n_{\mathbf{p}'_1} n_{\mathbf{p}'_2} (n_{\mathbf{k}} + n_{\mathbf{p}_1}))] \}. \end{aligned} \quad (5)$$

The first term in the right-hand-side of Eq. (5) is Hartree–Fock term which is first order in U . By taking $\mathbf{k} = \mathbf{k}'$, we can simplify Eq. (5):

$$\begin{aligned} \frac{dn_{\mathbf{k}}}{dt} &= -\frac{4\pi}{\hbar^2} \sum_{\mathbf{p}_1, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{k} - \mathbf{p}_1) U(\mathbf{p}'_2 - \mathbf{p}'_1) \{ 2\delta(\omega_{\mathbf{k}} + \omega_{\mathbf{p}_1} - \omega_{\mathbf{p}'_1} - \omega_{\mathbf{p}'_2}) \} \\ &\quad \times [n_{\mathbf{k}} n_{\mathbf{p}_1} - n_{\mathbf{p}'_1} n_{\mathbf{p}'_2} + \eta (n_{\mathbf{k}} n_{\mathbf{p}_1} (n_{\mathbf{p}'_1} + n_{\mathbf{p}'_2}) - n_{\mathbf{p}'_1} n_{\mathbf{p}'_2} (n_{\mathbf{k}} + n_{\mathbf{p}_1}))], \end{aligned} \quad (6)$$

where the Hartree–Fock contribution vanishes identically and will be absorbed into a renormalized single-particle dispersion, or, in a form that shows enhancement/blocking due to stimulated scattering

$$\begin{aligned} \frac{dn_{\mathbf{k}}}{dt} &= -\frac{8\pi}{\hbar^2} \sum_{\mathbf{p}_1, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{k} - \mathbf{p}_1) U(\mathbf{p}'_2 - \mathbf{p}'_1) \{ \delta(\omega_{\mathbf{k}} + \omega_{\mathbf{p}_1} - \omega_{\mathbf{p}'_1} - \omega_{\mathbf{p}'_2}) \} \\ &\quad [n_{\mathbf{k}} n_{\mathbf{p}_1} (1 + \eta n_{\mathbf{p}'_1}) (1 + \eta n_{\mathbf{p}'_2}) - n_{\mathbf{p}'_1} n_{\mathbf{p}'_2} (1 + \eta n_{\mathbf{k}}) (1 + \eta n_{\mathbf{p}_1})]. \end{aligned} \quad (7)$$

(The difference between the forms of Eqs. (6) and (7) lies in their antisymmetry under relabeling of dummy variables and therefore vanishes upon summation in the collision integral).

While the operator formulation renders the gain–loss structure of the collision integral immediately transparent, it does not provide a systematic organization of correlations and memory effects. In particular, irreversibility is not an intrinsic feature of the operator algebra, but emerges only after the introduction of additional approximations. These aspects are more directly addressed within the nonequilibrium Green-function framework, which we review in the following section.

3. Nonequilibrium Green-Function Derivation of QBE

In this section, we rederive the QBE within the NEGF framework developed by Kadanoff–Baym and Keldysh. The structure of the derivation closely parallels the ladder-operator construction presented in [Sec. 2](#). Specifically, we begin from the exact equations of motion for the one- and two-particle Green functions, apply Wick’s theorem to express the two-particle Green function in terms of products of one-particle Green functions, and reorganize the resulting contributions into the second-order (Born) self-energy using the Langreth–Wilkins projection rules. We then perform a

Wigner transformation of the Kadanoff–Baym kinetic equation and impose the quasiparticle approximation to obtain the Uehling–Uhlenbeck collision integral.

3.1. Kadanoff–Baym formalism

Following Kadanoff and Baym [5], we define the contour-ordered one- and two-particle Green functions

$$G(1, 1') = \frac{1}{i} \langle T_C \psi(1) \psi^\dagger(1') \rangle, \quad (8)$$

$$G_2(12; 1'2') = \frac{1}{i^2} \langle T_C \psi(1) \psi(2) \psi^\dagger(2') \psi^\dagger(1') \rangle, \quad (9)$$

where $1 \equiv (\mathbf{r}_1, t_1)$ and T_C orders operators along the Keldysh contour. The exact equations of motion for G read

$$\left(i\partial_{t_1} + \frac{\nabla_1^2}{2m} \right) G(1, 1') = \delta(1 - 1') + \eta i \int d\mathbf{r}_2 v(\mathbf{r}_1 - \mathbf{r}_2) G_2(12; 1'2^+) |_{t_2=t_1}, \quad (10)$$

$$\left(-i\partial_{t'_1} + \frac{\nabla_{1'}^2}{2m} \right) G(1, 1') = \delta(1 - 1') + \eta i \int d\mathbf{r}_2 G_2(12^-; 1'2) v(\mathbf{r}_2 - \mathbf{r}_{1'}), \quad (11)$$

which are the Green-function analog of the Heisenberg equation of motion for $\hat{a}_k^\dagger \hat{a}_k$ used in the ladder derivation (see Sec. 2). To close Eq. (10), it is necessary to approximate the two-particle Green function $G_2(12; 1'2')$. At second order in the interaction, this is achieved through Wick factorization, which reduces the six-field correlator to products of one-particle Green functions. Upon projection of the contour indices onto Keldysh components, this procedure generates twelve distinct real-time contractions that together form the second-order (Born) correlation self-energy.

The appearance of twelve correlation (gain–loss) channels is specific to the quartic, number-conserving interaction Hamiltonian considered here and follows directly from the Wick factorization of the six-field correlator at second order. More general interaction Hamiltonians, including those that do not conserve particle number, give rise to different kinetic structures and require corresponding extensions of the collision integral.

3.2. Wick factorization of G_2

For noninteracting fields, Wick’s theorem [3] implies the factorization of contour-ordered two-particle Green function into products of single-particle Green functions:

$$G_2(12; 1'2') = G(1, 1')G(2, 2') + \eta G(1, 2')G(2, 1'). \quad (12)$$

For four contour-ordered free fields, Wick’s theorem gives the two possible pairings of annihilation with creation operators:

$$\begin{aligned} \langle T_C \psi(1) \psi(2) \psi^\dagger(2') \psi^\dagger(1') \rangle &= \langle T_C \psi(1) \psi^\dagger(1') \rangle \langle T_C \psi(2) \psi^\dagger(2') \rangle \\ &+ \eta \langle T_C \psi(1) \psi^\dagger(2') \rangle \langle T_C \psi(2) \psi^\dagger(1') \rangle. \end{aligned} \quad (13)$$

By using the Kadanoff–Baym convention $\langle T_C \psi(a) \psi^\dagger(b) \rangle = i G(a, b)$, Eq. (13) becomes

$$\langle T_C \psi(1) \psi(2) \psi^\dagger(2') \psi^\dagger(1') \rangle = i^2 [G(1, 1') G(2, 2') + \eta G(1, 2') G(2, 1')], \quad (14)$$

which is Kadanoff–Baym Eqs. (3)–(25). At the operator level, the second-order Born self-energy involves a six-field correlator of the form $\langle T_C \psi(1) \psi(2) \psi(2') \psi^\dagger(1') \psi^\dagger(2) \psi^\dagger(2') \rangle$. A correlator involving six field operators admits fifteen distinct Wick pairings, consistent with the general combinatorial result $(2n - 1)!! = (6 - 1)!! = 15$. Of these, three pairings contract the external fields $\psi(1)$ and $\psi^\dagger(1')$ with each other and generate Hartree–Fock (mean-field) contributions, denoted collectively by Σ_{HF} . The remaining twelve pairings correspond to genuine correlation (Born) topologies. Upon implementing contour time ordering and performing the $<, >$ projections, these twelve Wick pairings give rise to twelve distinct real-time Born correlation channels. At the level of one-particle quantities, these channels reorganize naturally into two three-line structures:

$$X(1, 1') = G(1, 1') G(2, 2') G(2', 2), \quad Y(1, 1') = G(1, 2') G(2, 1') G(2', 2), \quad (15)$$

which may be identified as the direct and exchange contributions to the second-order Born self-energy. The associated Keldysh-contour representations are shown in Fig. 1, serving as a real-time bookkeeping device for the ordering of the three Green functions. Each topology generates three lesser and three greater real-time components; taken together, they account for the twelve collision channels explicitly enumerated in the ladder-operator derivation. The Hartree–Fock contributions are retained only insofar as they renormalize the single-particle dispersion and produce mean-field forces in the Liouville operator. As they do not contribute to irreversible scattering processes, they do not appear in the collision integral derived below.

3.3. Born self-energy via Langreth rules

We begin with Kadanoff–Baym’s decomposition of the self-energy

$$\Sigma(1, 1') = \Sigma_{\text{HF}}(1, 1') + \Sigma_c(1, 1'), \quad (16)$$

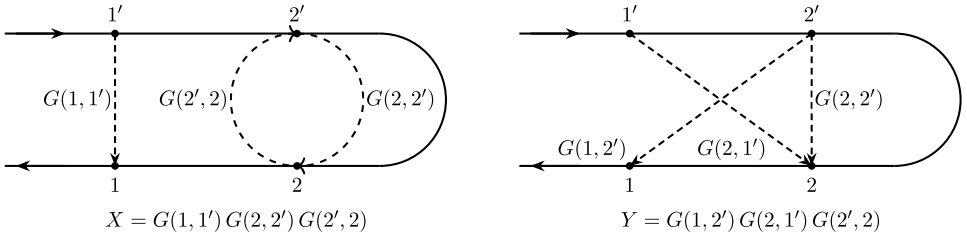


Fig. 1. Keldysh contour sketches for the direct (X) and exchange (Y) channels at Born level. The contour representation serves solely as a real-time bookkeeping device for implementing time ordering and projecting contour-ordered products onto their lesser and greater components.

where the Hartree–Fock term is local in time:

$$\Sigma_{\text{HF}}(1, 1') = \delta(t_1 - t'_1) \left\{ \delta(\mathbf{r}_1 - \mathbf{r}_{1'}) \int d\mathbf{r}_2 v(\mathbf{r}_1 - \mathbf{r}_2) \langle n(\mathbf{r}_2) \rangle + \eta i v(\mathbf{r}_1 - \mathbf{r}_2) G^<(1, 2) \Big|_{t_1=t_2} \right\} \quad (17)$$

and the correlation part at second order (Born approximation) reads

$$\begin{aligned} \Sigma_c(1, 1') = & \pm i^2 \int d\mathbf{r}_2 d\mathbf{r}_{2'} v(\mathbf{r}_1 - \mathbf{r}_2) v(\mathbf{r}_{1'} - \mathbf{r}_{2'}) [G(1, 1') G(2, 2') G(2', 2) \\ & + \eta G(1, 2') G(2, 1') G(2', 2)]_{t_2=t_1, t'_2=t'_1}. \end{aligned} \quad (18)$$

To connect with real-time kinetic theory, we project the contour products in Eq. (18) onto Keldysh components using the Langreth–Wilkins theorem [7]. For a contour product of three functions, one has

$$(ABC)^< = A^R B^R C^< + A^R B^< C^A + A^< B^A C^A, \quad (19)$$

with an analogous expression for $(ABC)^>$. We apply Eq. (19) to the direct structure equation (15) at $t_2 = t_1$ and $t_{2'} = t_{1'}$, we obtain

$$\begin{aligned} X^<(1, 1') = & G^R(1, 1') G^R(2, 2') G^<(2', 2) + G^R(1, 1') G^<(2, 2') G^A(2', 2) \\ & + G^<(1, 1') G^A(2, 2') G^A(2', 2), \end{aligned} \quad (20)$$

$$\begin{aligned} Y^<(1, 1') = & G^R(1, 2') G^R(2, 1') G^<(2', 2) + G^R(1, 2') G^<(2, 1') G^A(2', 2) \\ & + G^<(1, 2') G^A(2, 1') G^A(2', 2). \end{aligned} \quad (21)$$

Similarly, we can write for $X^>(1, 1')$ and $Y^>(1, 1')$. In the Born and quasiparticle limits, where spectral functions are narrow and memory effects negligible, the dominant contribution to the collision kernel arises from terms containing one lesser and two greater components. Upon imposing the on-shell (quasiparticle) approximation, enforcing energy conservation through the Markov limit and neglecting vertex and memory corrections, this contribution reduces to the Uehling–Uhlenbeck collision structure. We apply the Langreth–Wilkins projection rules to each term in the Kadanoff–Baym equations prior to taking equal-time limits. After projection, we impose the constraints $t_2 = t_1$ and $t'_2 = t'_{1'}$, ensuring that all kinetic quantities are evaluated at a single macroscopic time. Retaining only the on-shell contributions then yields the direct term of the lesser self-energy:

$$\Sigma_{\text{c, dir}}^<(1, 1') = \pm i^2 \int d2 d2' v(1 - 2) v(1' - 2') G^<(2, 2') G^>(1, 2') G^>(2, 1') \Big|_{t_2=t_1, t_{2'}=t_{1'}}. \quad (22)$$

Keeping again only the collision-relevant “one lesser, two greater” terms gives

$$\Sigma_{\text{c, ex}}^<(1, 1') = \pm i^2 \int d2 d2' v(1 - 2) v(1' - 2') G^<(2, 1') G^>(1, 2') G^>(2', 2) \Big|_{t_2=t_1, t_{2'}=t_{1'}}. \quad (23)$$

Counting the contributions from the lesser and greater components, $X^<, X^>, Y^<, Y^>$, one obtains a total of $3 + 3 + 3 + 3 = 12$ distinct real-time terms. These contributions are in one-to-one correspondence with the twelve Wick contractions of the six-field correlator that defines the second-order Born self-energy. Accordingly, the components of the Born self-energy can be expressed in terms of these twelve real-time contributions as follows:

$$\Sigma_c^<(1, 1') = \pm i^2 \int d2 d2' v(1-2)v(1'-2') [X^<(1, 1') + \eta Y^<(1, 1')], \quad (24)$$

$$\Sigma_c^>(1, 1') = \pm i^2 \int d2 d2' v(1-2)v(1'-2') [X^>(1, 1') + \eta Y^>(1, 1')]. \quad (25)$$

At this stage, the dynamics remains governed by the full Kadanoff–Baym equations, which retain explicit two-time memory integrals and are fully consistent with microscopic reversibility. As emphasized in standard treatments of quantum kinetic theory, the emergence of a Boltzmann-type kinetic equation requires additional controlled approximations beyond Wick factorization alone. Combining Eqs. (22) and (23), we obtain the explicit coordinate-space Born expression for the lesser correlation self-energy:

$$\begin{aligned} \Sigma_c^<(1, 1') = & \pm i^2 \int d2 d2' v(1-2)v(1'-2') [G^<(2, 2') G^>(1, 2') G^>(2, 1') \\ & \pm G^<(2, 1') G^>(1, 2') G^>(2', 2)]|_{t_2=t_1, t_{2'}=t_{1'}}. \end{aligned} \quad (26)$$

An analogous calculation (interchanging $<$ and $>$) gives the greater correlation self-energy. The lesser and greater correlation self-energy components constitute an intermediate level of description. They follow directly from Eqs. (16)–(18), but are expressed explicitly in terms of real-time Keldysh components. In this form, they constitute the Green-function counterparts of the twelve Wick contraction channels identified in the ladder-operator derivation presented in Sec. 2. Substituting these self-energies into the Kadanoff–Baym kinetic equation for $G^<(1, 1')$ yields the Born collision integral of the quantum Boltzmann equation after performing the Wigner transformation and imposing the quasiparticle approximation.

This point was analyzed in detail by Danielewicz, who showed that the Boltzmann equation emerges only after invoking both a weak-coupling Born approximation a Markovian coarse-graining in time, and a quasiparticle projection onto energy shells [10].

3.4. Wigner transform and gradient

To derive a kinetic equation, we introduce the Wigner center-of-mass and relative coordinates [11], defined by $(\mathbf{R}, T) = (x_1 + x_2)/2$ and $(\mathbf{r}, t) = x_1 - x_2$ with

$$x_1 = \left(T + \frac{t}{2}, \mathbf{R} + \frac{\mathbf{r}}{2}\right), \quad x_2 = \left(T - \frac{t}{2}, \mathbf{R} - \frac{\mathbf{r}}{2}\right). \quad (27)$$

In terms of these variables, the lesser Green function takes the following form:

$$G^<(\mathbf{r}, t; \mathbf{R}, T) = i \left\langle \psi^\dagger \left(\mathbf{R} - \frac{\mathbf{r}}{2}, T - \frac{t}{2} \right) \psi \left(\mathbf{R} + \frac{\mathbf{r}}{2}, T + \frac{t}{2} \right) \right\rangle \quad (28)$$

and its Wigner transform is

$$G^<(\mathbf{R}, T; \mathbf{p}, \omega) = \int d^3r \int dt e^{i(\omega t - \mathbf{p} \cdot \mathbf{r})} G^<(\mathbf{r}, t; \mathbf{R}, T). \quad (29)$$

In the notation of Mahan, the semiclassical Boltzmann distribution $f(\mathbf{R}, \mathbf{v}, T)$ is obtained as the frequency integral of the corresponding Wigner distribution function $f(\mathbf{R}, T; \mathbf{p}, \omega)$, i.e.,

$$f(\mathbf{R}, \mathbf{v}, T) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} f(m\mathbf{v}, \omega; \mathbf{R}, T). \quad (30)$$

Within the ladder-operator formulation, the Wigner distribution is defined as

$$f(t, \mathbf{r}; \mathbf{p}, \omega) = \int d\mathbf{r}' e^{-i\mathbf{p} \cdot \mathbf{r}'} \int dt' e^{i\omega t'} \left\langle \hat{\Psi}^\dagger \left(t - \frac{t'}{2}, \mathbf{r} - \frac{\mathbf{r}'}{2} \right) \hat{\Psi} \left(t + \frac{t'}{2}, \mathbf{r} + \frac{\mathbf{r}'}{2} \right) \right\rangle. \quad (31)$$

Comparing with the Kadanoff–Baym/Mahan lesser Green function and identifying $(\mathbf{R}, T) \rightarrow (\mathbf{r}, t)$, we obtain

$$f(t, \mathbf{r}; \mathbf{p}, \omega) = \frac{1}{i} \int d\mathbf{r}' dt' e^{i(\omega t' - \mathbf{p} \cdot \mathbf{r}')} G^<(\mathbf{r}', t'; \mathbf{r}, t) = \frac{1}{i} G^<(\mathbf{r}, t; \mathbf{p}, \omega), \quad (32)$$

so that the phase-space distribution

$$f(t, \mathbf{r}, \mathbf{p}) \equiv \int \frac{d\omega}{2\pi} f(t, \mathbf{r}; \mathbf{p}, \omega) = -i \int \frac{d\omega}{2\pi} G^<(\mathbf{r}, t; \mathbf{p}, \omega) \quad (33)$$

is precisely the frequency integral of the Wigner-transformed lesser Green function, in agreement with Mahan's [11, Eq. (3.337)]. Subtracting Eq. (11) from Eq. (10) and projecting onto the lesser component then yields the exact kinetic equation:

$$\left(i\partial_{t_1} + i\partial_{t_{1'}} + \frac{\nabla_1^2 - \nabla_{1'}^2}{2m} \right) G^<(1, 1') = \int d2 [\Sigma^>(1, 2) G^<(2, 1') - \Sigma^<(1, 2) G^>(2, 1')]. \quad (34)$$

Performing a gradient expansion to leading order in space and time variations yields the KB kinetic equation (34) in Wigner representation:

$$\partial_t G^<(\mathbf{r}, t; \mathbf{p}, \omega) + \frac{\mathbf{p}}{m} \cdot \nabla_{\mathbf{r}} G^<(\mathbf{r}, t; \mathbf{p}, \omega) = \Sigma^>(\mathbf{r}, t; \mathbf{p}, \omega) G^< - \Sigma^<(\mathbf{r}, t; \mathbf{p}, \omega) G^>, \quad (35)$$

where products and arguments $(\mathbf{r}, t; \mathbf{p}, \omega)$ are implicit. Decomposing the self-energy as

$$\begin{aligned} \Sigma(\mathbf{r}, t; \mathbf{p}, \omega) &= \Sigma_{\text{HF}}(\mathbf{r}, t; \mathbf{p}) + \Sigma_{\text{coll}}(\mathbf{r}, t; \mathbf{p}, \omega), \\ \Sigma_{\text{HF}}(\mathbf{r}, t; \mathbf{p}) &= \sum_{\mathbf{q}} U(\mathbf{p} - \mathbf{q}) n_{\mathbf{q}}(\mathbf{r}, t), \end{aligned} \quad (36)$$

one obtains the full KB–Wigner kinetic equation:

$$\begin{aligned} & \left(\partial_t + \frac{\mathbf{p}}{m} \cdot \nabla_{\mathbf{r}} - (\nabla_{\mathbf{r}} \Sigma_{\text{HF}}) \cdot \nabla_{\mathbf{p}} \right) G^<(\mathbf{r}, t; \mathbf{p}, \omega) \\ & = \Sigma_{\text{coll}}^>(\mathbf{r}, t; \mathbf{p}, \omega) G^< - \Sigma_{\text{coll}}^<(\mathbf{r}, t; \mathbf{p}, \omega) G^>, \end{aligned} \quad (37)$$

which is the NEGF analogue of the Liouville operator plus a collision term in the ladder-operator QBE.

3.5. Quasiparticle ansatz and on-shell reduction

To obtain a kinetic equation for the occupation number $n_{\mathbf{p}}$, we now invoke the quasiparticle (on-shell) approximation. Therefore, the full Green functions are projected onto delta functions supported on the dispersion $\omega = \omega_{\mathbf{p}}$, which allows the Born self-energy in Eq. (26) and its analog to be evaluated explicitly at the quasiparticle pole.

Assuming a controlled separation between microscopic and macroscopic time scales, the two-time Kadanoff–Baym equations can then be reduced to a Markovian kinetic equation by coarse-graining the memory integrals over relative times. This procedure removes explicit temporal nonlocality and enforces energy conservation in the long-time limit. It is important to emphasize, however, that this reduction is not a formal identity, its validity relies on weak coupling and on the rapid decay of correlation functions. Adopting the Keldysh convention ($G^< = i\langle\psi^\dagger\psi\rangle$, $G^> = -i\langle\psi\psi^\dagger\rangle$), the quasiparticle approximation for a weakly interacting gas is

$$G^<(\mathbf{p}, \omega) = 2\pi i \delta(\omega - \omega_{\mathbf{p}}) n_{\mathbf{p}}, \quad G^>(\mathbf{p}, \omega) = -2\pi i \delta(\omega - \omega_{\mathbf{p}}) (1 + \eta n_{\mathbf{p}}), \quad (38)$$

where $\omega_{\mathbf{p}} = \mathbf{p}^2/2m$ and $n_{\mathbf{p}}$ is the mode occupation number. The quasiparticle approximation underlying Eq. (38) projects the dynamics onto sharp energy shells. As shown in detailed analyses of off-shell transport, this step eliminates spectral broadening and memory effects, rendering the kinetic equation local in time at the cost of discarding off-shell coherence [5, 9]. We define the momentum occupation number in the Green-function language by

$$n_{\mathbf{p}}(t) \equiv -i \int \frac{d\omega}{2\pi} G^<(\mathbf{p}, \omega, t), \quad (39)$$

which reduces to the ladder-operator definition $n_{\mathbf{p}} = \langle \hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}} \rangle$ in a homogeneous system. Multiplying Eq. (34) by $(-i)$ and integrating over ω gives

$$\frac{dn_{\mathbf{p}}}{dt} = \int \frac{d\omega}{2\pi} [\Sigma^>(\mathbf{p}, \omega) G^<(\mathbf{p}, \omega) - \Sigma^<(\mathbf{p}, \omega) G^>(\mathbf{p}, \omega)], \quad (40)$$

which is the Green-function counterpart of Eq. (4) in the ladder derivation. We now evaluate the lesser and greater Born correlation self-energy $\Sigma_{\text{coll}}^{\lessgtr}$ in momentum representation using Eq. (26) and its analog equation, respectively. For a translationally invariant interaction, the potential depends only on the relative coordinate and admits

the Fourier expansion:

$$v(\mathbf{r}_1 - \mathbf{r}_2) = \frac{1}{V} \sum_{\mathbf{q}} U(\mathbf{q}) e^{i\mathbf{q} \cdot (\mathbf{r}_1 - \mathbf{r}_2)}. \quad (41)$$

Substituting Eq. (41) and the plane-wave expansions of the Green functions into the coordinate-space expression for $\Sigma_c^<(1, 1')$ and carrying out the spatial integrations produces momentum-conserving delta functions and the characteristic pair of interaction factors

$$U(\mathbf{p} - \mathbf{p}_1) U(\mathbf{p}'_2 - \mathbf{p}'_1). \quad (42)$$

For a Hermitian interaction $v(\mathbf{r})$, one has $U(-\mathbf{q}) = U^*(\mathbf{q})$, and momentum conservation implies $\mathbf{q} \equiv \mathbf{p} - \mathbf{p}_1 = \mathbf{p}'_2 - \mathbf{p}'_1$. The product then reduces to

$$U(\mathbf{p} - \mathbf{p}_1) U(\mathbf{p}'_2 - \mathbf{p}'_1) = U(\mathbf{q}) U(-\mathbf{q}) = |U(\mathbf{q})|^2, \quad (43)$$

which is precisely the squared scattering amplitude obtained from Sec. 2. Using the quasiparticle forms and the momentum structure (42), the Born self-energies evaluated on shell simplify to

$$\begin{aligned} \Sigma_c^<(\mathbf{p}, \omega_{\mathbf{p}}) &= (2\pi) \sum_{\mathbf{p}_1, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{p} - \mathbf{p}_1) U(\mathbf{p}'_2 - \mathbf{p}'_1) \delta(\omega_{\mathbf{p}} + \omega_{\mathbf{p}_1} - \omega_{\mathbf{p}'_1} - \omega_{\mathbf{p}'_2}) \\ &\quad \times n_{\mathbf{p}_1} (1 + \eta n_{\mathbf{p}'_1}) (1 + \eta n_{\mathbf{p}'_2}), \end{aligned} \quad (44)$$

$$\begin{aligned} \Sigma_c^>(\mathbf{p}, \omega_{\mathbf{p}}) &= (2\pi) \sum_{\mathbf{p}_1, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{p} - \mathbf{p}_1) U(\mathbf{p}'_2 - \mathbf{p}'_1) \delta(\omega_{\mathbf{p}} + \omega_{\mathbf{p}_1} - \omega_{\mathbf{p}'_1} - \omega_{\mathbf{p}'_2}) \\ &\quad \times (1 + \eta n_{\mathbf{p}_1}) n_{\mathbf{p}'_1} n_{\mathbf{p}'_2}. \end{aligned} \quad (45)$$

Inserting Eqs. (44) and (45) into the kinetic equation (40), and relabeling the external momentum $\mathbf{p} \rightarrow \mathbf{k}$ to match Sec. 2, we arrive at

$$\begin{aligned} \frac{dn_{\mathbf{k}}}{dT} &= \sum_{\mathbf{p}_1, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{k} - \mathbf{p}_1) U(\mathbf{p}'_2 - \mathbf{p}'_1) 2\pi \delta(\omega_{\mathbf{k}} + \omega_{\mathbf{p}_1} - \omega_{\mathbf{p}'_1} - \omega_{\mathbf{p}'_2}) \\ &\quad \times [(1 + \eta n_{\mathbf{k}})(1 + \eta n_{\mathbf{p}_1}) n_{\mathbf{p}'_1} n_{\mathbf{p}'_2} - n_{\mathbf{k}} n_{\mathbf{p}_1} (1 + \eta n_{\mathbf{p}'_1})(1 + \eta n_{\mathbf{p}'_2})], \end{aligned} \quad (46)$$

where T describes the coarse-grained kinetic time variable (This time scale allows fast microscopic oscillations to be averaged out while retaining irreversible population dynamics). Restoring the Fourier-transform normalizations and factors of $\hbar$ yields the final Uehling–Uhlenbeck quantum Boltzmann equation

$$\begin{aligned} \frac{dn_{\mathbf{k}}}{dT} &= -\frac{8\pi}{\hbar^2} \sum_{\mathbf{p}_1, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{k} - \mathbf{p}_1) U(\mathbf{p}'_2 - \mathbf{p}'_1) \delta(\omega_{\mathbf{k}} + \omega_{\mathbf{p}_1} - \omega_{\mathbf{p}'_1} - \omega_{\mathbf{p}'_2}) \\ &\quad \times [n_{\mathbf{k}} n_{\mathbf{p}_1} (1 + \eta n_{\mathbf{p}'_1})(1 + \eta n_{\mathbf{p}'_2}) - n_{\mathbf{p}'_1} n_{\mathbf{p}'_2} (1 + \eta n_{\mathbf{k}})(1 + \eta n_{\mathbf{p}_1})], \end{aligned} \quad (47)$$

identical in structure to the ladder-operator result, Eq. (7). Every gain and loss term, every Bose enhancement / Pauli blocking factor, and every $|U(\mathbf{k} - \mathbf{p}_1)|^2$ scattering kernel can thus be traced back to a specific Wick contraction and its corresponding

Langreth-projected Born self-energy contribution. Prior to taking the Markov limit, the collision term retains a nonlocal dependence on the system's history, reflecting the finite duration of microscopic correlations. This non-Markovian structure is an intrinsic feature of the Kadanoff–Baym equations and has been emphasized in modern treatments of quantum kinetic theory. Irreversibility and locality in time emerge only after invoking a separation of time scales between fast microscopic correlation dynamics and slow macroscopic evolution of the distribution function, an assumption that underlies the Markov approximation [12]. This correspondence makes explicit that the operator contraction channels and the Born self-energy components encode the same kinetic content.

4. Discussion

This section interprets the microscopic derivation in terms of irreversibility, entropy production and its relevance to established kinetic frameworks.

4.1. Irreversibility and entropy production

From the perspective of the correlation hierarchy, the quantum Boltzmann equation corresponds to a controlled truncation of the Bogoliubov–Born–Green–Kirkwood–Yvon (BBGKY) hierarchy at the two-particle level, together with a factorization assumption for higher-order correlations. The resulting autonomy of the kinetic equation reflects not a failure of microscopic reversibility, but a deliberate loss of information about higher-order degrees of freedom that renders the reduced description closed.

It is important to emphasize that neither Wick factorization nor the Born approximation alone introduces irreversibility. The Kadanoff–Baym equations remain time-reversal invariant as long as their full two-time structure is retained. Irreversibility enters only after the Markov approximation and the quasiparticle projection, which together discard phase information, suppress memory effects and truncate the correlation hierarchy. Microscopic reversibility, entropy production and macroscopic irreversibility are therefore distinct notions, which only coincide after kinetic coarse-graining.

The H-theorem does not assert a fundamental arrow of time, but instead quantifies the monotonic increase of coarse-grained kinetic entropy within a reduced description that has discarded phase information and higher-order correlations [13, 14]. The appearance of irreversibility is therefore a property of the kinetic approximation, not of the underlying unitary dynamics.

4.2. Quantum plasma physics

The quantum Boltzmann equation derived in this review provides the microscopic kinetic foundation for a wide class of quantum plasma theories. In plasma physics, the same Markovian, quasiparticle quantum Boltzmann equation underlies the description of collisional relaxation, transport coefficients and entropy production in weakly coupled quantum plasmas.

In particular, the kinetic equations employed in quantum plasma analyses such as [15] are obtained by specializing the general quantum Boltzmann framework to long-range Coulomb interactions, incorporating screening and collective effects appropriate to plasmas. While such works focus on plasma-specific physics, including dielectric response and transport, they typically assume the structure of the collision integral as given.

In plasma applications, the interaction entering the collision integral is dynamically screened by collective effects, leading to dielectric or Lenard–Balescu-type modifications of the bare Coulomb potential, without altering the underlying Born–Markov kinetic structure derived here [16–18].

The present derivation complements quantum plasma approaches by making explicit the microscopic origin of the gain–loss structure, through a direct mapping between the twelve Wick contraction channels and the lesser and greater Born self-energies.

5. Conclusion

This review has provided a unified synthesis of the microscopic foundations of the quantum Boltzmann equation, bridging operator and nonequilibrium Green-function formulations within a single transparent framework.

Beyond the Markovian limit treated here, quantum kinetic theory admits systematic extensions that retain memory effects, off-shell dynamics and strong-coupling correlations. Non-Markovian collision integrals arise naturally at the level of the full two-time Kadanoff–Baym equations, while ladder and T -matrix resummations capture resonant scattering, bound-state formation and transport beyond the dilute quasiparticle regime. Such approaches are essential near criticality, in low-dimensional systems, and in modern cold-atom and strongly correlated condensed-matter settings, where the assumptions underlying the quantum Boltzmann equation break down.

In this work, we have provided an explicit microscopic bridge between the ladder-operator formulation of the quantum Boltzmann equation and its nonequilibrium Green-function representation. Wick contractions determine the algebraic structure of scattering amplitudes, while irreversibility enters only through subsequent kinetic approximations. By carrying out the full Wick expansion of the six-field correlator and tracking the complete Wick contraction structure through the Kadanoff–Baym equations, we have made transparent how the Born self-energy, the Wigner transform, and the on-shell quasiparticle reduction collectively generate the Uehling–Uhlenbeck collision integral. Under the Born, Markov and quasiparticle approximations, the quantum Boltzmann equation follows systematically from the Kadanoff–Baym equations, and these same approximations uniquely determine the structure of the collision integral.

The result is a unified operator-Green-function derivation of the quantum Boltzmann equation, in which every gain and loss term, every Bose or Pauli factor and every $|U(\mathbf{k} - \mathbf{p})|^2$ scattering kernel arises transparently from an explicit

contraction structure. Within the domain defined by Wick's theorem, the Born–Markov approximation, and the applicability of Keldysh theory, this work provides a systematic microscopic reconstruction of standard quantum kinetic theory.

Phase-space and Keldysh approaches provide complementary perspectives on this reduction, with stochasticity emerging naturally once higher-order correlations are integrated out [19]. From this viewpoint, alternative formulations such as Boltzmann–Langevin or stochastic kinetic theories describe fluctuations around the mean quantum Boltzmann evolution, emphasizing noise correlations and fluctuation–dissipation relations rather than the deterministic gain–loss structure itself. In this work, such noise is not introduced phenomenologically but is understood as an emergent consequence of coarse-graining and correlation truncation within the underlying operator and nonequilibrium Green-function framework. No additional noise model is assumed here; rather, stochasticity appears as a controlled consequence of integrating out higher-order correlations within the microscopic theory.

Finally, the same Born self-energies that generate the quantum Boltzmann collision integral also control equilibrium thermodynamic quantities, underscoring the deep connection between quantum kinetics and statistical mechanics. By making the operator-level contractions and the Green-function structure fully explicit, this review provides a transparent and systematic reconstruction of the microscopic steps leading to the quantum Boltzmann equation.

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Appendix A. Explicit Wick Contraction Algebra

This appendix provides the explicit algebraic details underlying the Wick-contraction analysis summarized in Sec. 2; these steps are not required for following the main conceptual argument but are included for completeness and transparency. As a result, we obtain

$$\begin{aligned} \frac{d}{dt}(\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}'}) &= \frac{i}{2\hbar} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{p}_1 - \mathbf{p}_2) \left[\hat{a}_{\mathbf{p}'_1}^\dagger \hat{a}_{\mathbf{p}'_2}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_2 \mathbf{k}} + \hat{a}_{\mathbf{p}'_1}^\dagger \hat{a}_{\mathbf{p}'_2}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_1 \mathbf{k}} \right. \\ &\quad \left. - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}'_2}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}'_1 \mathbf{k}'} - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}'_1}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}'_2 \mathbf{k}'} \right] - \frac{1}{4\hbar^2} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}'_1, \mathbf{p}'_2} \sum_{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}'_1, \mathbf{q}'_2} \\ &\quad \times U(\mathbf{p}_1 - \mathbf{p}_2) U(\mathbf{q}_1 - \mathbf{q}_2) \int_0^t dt' \left[\hat{a}_{\mathbf{q}'_1}^\dagger \hat{a}_{\mathbf{q}'_2}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}'_1}^\dagger \hat{a}_{\mathbf{p}'_2}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_2 \mathbf{k}} \right. \\ &\quad \left. - \hat{a}_{\mathbf{p}'_1}^\dagger \hat{a}_{\mathbf{p}'_2}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}'_1}^\dagger \hat{a}_{\mathbf{q}'_2}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{p}_2 \mathbf{k}} + \hat{a}_{\mathbf{q}'_1}^\dagger \hat{a}_{\mathbf{q}'_2}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}'_1}^\dagger \hat{a}_{\mathbf{p}'_2}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_1 \mathbf{k}} \right] \end{aligned}$$

$$\begin{aligned}
& -\hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{p}_1 \mathbf{k}} - \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_1' \mathbf{k}'} \\
& + \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{p}_1' \mathbf{k}'} - \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_2' \mathbf{k}'} \\
& + \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{p}_2' \mathbf{k}'} \Big].
\end{aligned} \tag{A.1}$$

In the interaction picture, operators evolve as $a_{\mathbf{p}}(t) = a_{\mathbf{p}} e^{-i\omega_{\mathbf{p}} t}$, so each term in the second-order expansion contains oscillatory factors of the form $e^{-i(\omega_{\mathbf{p}_1'} + \omega_{\mathbf{p}_2'} - \omega_{\mathbf{k}'} - \omega_{\mathbf{p}_1})\tau}$. The time integral therefore takes the standard Born–Markov form:

$$\begin{aligned}
& \int_0^\infty d\tau e^{i(\omega_{\mathbf{p}_1'} + \omega_{\mathbf{p}_2'} - \omega_{\mathbf{k}'} - \omega_{\mathbf{p}_1})\tau} \\
& = \pi \delta(\omega_{\mathbf{p}_1'} + \omega_{\mathbf{p}_2'} - \omega_{\mathbf{k}'} - \omega_{\mathbf{p}_1}) + i \mathcal{P} \left(\frac{1}{\omega_{\mathbf{p}_1'} + \omega_{\mathbf{p}_2'} - \omega_{\mathbf{k}'} - \omega_{\mathbf{p}_1}} \right),
\end{aligned}$$

which enforces energy conservation in the long-time limit. This δ -function arises from the Markov approximation, not from an operator identity. All δ -functions arising from the commutators must be applied before Wick contractions are evaluated. Only after momenta are matched can diagonal contractions $\langle a_k^\dagger a_k \rangle = n_k$ and $\langle a_k a_k^\dagger \rangle = 1 + \eta n_k$ be inserted. As a result, evolution equation can be written as

$$\begin{aligned}
\frac{d}{dt} (\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}'}) &= \frac{i}{2\hbar} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_1', \mathbf{p}_2'} U(\mathbf{p}_1 - \mathbf{p}_2) \Big(\hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_2 \mathbf{k}} + \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{k}} \delta_{\mathbf{p}_1 \mathbf{k}} \\
& - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_1' \mathbf{k}'} - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_2' \mathbf{k}'} \Big) - \frac{\pi}{2\hbar^2} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_1', \mathbf{p}_2', \mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_1', \mathbf{q}_2'} \\
& \times U(\mathbf{p}_1 - \mathbf{p}_2) U(\mathbf{q}_1 - \mathbf{q}_2) \Big[\delta(\omega_{\mathbf{p}_1'} + \omega_{\mathbf{p}_2'} - \omega_{\mathbf{k}'} - \omega_{\mathbf{p}_1}) \\
& \times (\hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_2 \mathbf{k}} - \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{p}_2 \mathbf{k}}) \\
& + \delta(\omega_{\mathbf{p}_1'} + \omega_{\mathbf{p}_2'} - \omega_{\mathbf{p}_2} - \omega_{\mathbf{k}'})(\hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_1 \mathbf{k}} \\
& - \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{p}_1 \mathbf{k}}) - \delta(\omega_{\mathbf{k}} + \omega_{\mathbf{p}_2'} - \omega_{\mathbf{p}_2} - \omega_{\mathbf{p}_1}) \\
& \times (\hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_1' \mathbf{k}'} - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{p}_1' \mathbf{k}'} \\
& - \delta(\omega_{\mathbf{k}} + \omega_{\mathbf{p}_1'} - \omega_{\mathbf{p}_2} - \omega_{\mathbf{p}_1})(\hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_2' \mathbf{k}'} \\
& - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{p}_2' \mathbf{k}'} \Big].
\end{aligned} \tag{A.2}$$

The first and the second lines in [...] differ by interchange $\mathbf{p}_1 \leftrightarrow \mathbf{p}_2$ and, hence, due to symmetry $U(\mathbf{p}_1 - \mathbf{p}_2) = U(\mathbf{p}_2 - \mathbf{p}_1)$, they give the same contribution. Similarly, the

third and the forth lines give the same contribution. Therefore

$$\begin{aligned}
 \frac{d}{dt} \left(\hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{k}'} \right) = & \frac{i}{2\hbar} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}'_1, \mathbf{p}'_2} U(\mathbf{p}_1 - \mathbf{p}_2) \left[\hat{a}_{\mathbf{p}_1}^{\dagger} \hat{a}_{\mathbf{p}_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \delta_{\mathbf{p}_2 \mathbf{k}} + \hat{a}_{\mathbf{p}_1}^{\dagger} \hat{a}_{\mathbf{p}_2}^{\dagger} \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_1 \mathbf{k}} \right. \\
 & - \hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{p}_2}^{\dagger} \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_1 \mathbf{k}'} - \hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{p}_1}^{\dagger} \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_2 \mathbf{k}'} \left. \right] - \frac{\pi}{\hbar^2} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}'_1, \mathbf{p}'_2} \sum_{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}'_1, \mathbf{q}'_2} \\
 & \times U(\mathbf{p}_1 - \mathbf{p}_2) U(\mathbf{q}_1 - \mathbf{q}_2) \left\{ \delta(\omega_{\mathbf{p}'_1} + \omega_{\mathbf{p}'_2} - \omega_{\mathbf{k}'} - \omega_{\mathbf{p}_1}) \delta_{\mathbf{p}_2 \mathbf{k}} \right. \\
 & \times \left[\hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} - \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \right] \\
 & - \delta(\omega_{\mathbf{k}} + \omega_{\mathbf{p}'_1} - \omega_{\mathbf{p}_2} - \omega_{\mathbf{p}_1}) \delta_{\mathbf{p}_2 \mathbf{k}'} \left[\hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \right. \\
 & \left. \left. - \hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \right] \right\}. \quad (\text{A.3})
 \end{aligned}$$

By taking into account commutation relations (2.1) we obtain

$$\begin{aligned}
 & \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} - \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \\
 & = \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}'_1} \delta_{\mathbf{q}_1 \mathbf{p}'_2} + \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_1 \mathbf{p}'_1} \delta_{\mathbf{q}_2 \mathbf{p}'_2} \\
 & - \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}'_1 \mathbf{k}'} \delta_{\mathbf{q}'_2 \mathbf{p}_1} - \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}'_2 \mathbf{k}'} \delta_{\mathbf{q}'_1 \mathbf{p}_1} \\
 & + \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}'_1} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}'_2} + \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}'_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_1 \mathbf{p}'_2} \\
 & + \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}'_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}'_1} + \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}'_2} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_1 \mathbf{p}'_1} \\
 & - \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{q}'_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}'_2 \mathbf{k}'} - \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{q}'_2} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}'_1 \mathbf{k}'} \\
 & - \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{q}'_2} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}'_1 \mathbf{p}_1} - \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{q}'_1} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}'_2 \mathbf{p}_1}. \quad (\text{A.4})
 \end{aligned}$$

Due to symmetry of the sum with respect to interchange $\mathbf{q}_1 \leftrightarrow \mathbf{q}_2$, and $\mathbf{q}'_1 \leftrightarrow \mathbf{q}'_2$, we can replace

$$\begin{aligned}
 & \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \\
 & - \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \rightarrow 2 \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}'_1} \delta_{\mathbf{q}_1 \mathbf{p}'_2} \\
 & - 2 \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}'_1 \mathbf{k}'} \delta_{\mathbf{q}'_2 \mathbf{p}_1} + 2 \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}'_1} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}'_2} \\
 & + 2 \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}'_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}'_1} - 2 \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{q}'_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}'_2 \mathbf{k}'} \\
 & - 2 \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}'_2}^{\dagger} \hat{a}_{\mathbf{q}'_2} \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}'_1 \mathbf{p}_1}.
 \end{aligned}$$

Similarly

$$\begin{aligned}
 & \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \\
 & - \hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \rightarrow 2 \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \delta_{\mathbf{q}_2 \mathbf{p}'_1} \delta_{\mathbf{q}_1 \mathbf{k}} \\
 & - 2 \hat{a}_{\mathbf{p}'_1}^{\dagger} \hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}'_1 \mathbf{p}_2} \delta_{\mathbf{q}'_2 \mathbf{p}_1} + 2 \hat{a}_{\mathbf{q}'_1}^{\dagger} \hat{a}_{\mathbf{q}'_2}^{\dagger} \hat{a}_{\mathbf{p}'_1} \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \delta_{\mathbf{q}_2 \mathbf{k}}
 \end{aligned}$$

$$\begin{aligned}
& + 2\hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \delta_{\mathbf{q}_2 \mathbf{p}_1'} - 2\hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_2 \mathbf{p}_2} \\
& - 2\hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_1 \mathbf{p}_1}.
\end{aligned}$$

As a result, the evolution equation becomes

$$\begin{aligned}
\frac{d}{dt}(\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}'}) &= \frac{i}{2\hbar} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_1', \mathbf{p}_2'} U(\mathbf{p}_1 - \mathbf{p}_2) \left(\hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_2 \mathbf{k}} + \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{k}} \delta_{\mathbf{p}_1 \mathbf{k}} \right. \\
& - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_2}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_1' \mathbf{k}'} - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_2' \mathbf{k}'} \Big) - \frac{2\pi}{\hbar^2} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_1', \mathbf{p}_2'} \sum_{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_1', \mathbf{q}_2'} \\
& \times U(\mathbf{p}_1 - \mathbf{p}_2) U(\mathbf{q}_1 - \mathbf{q}_2) \left[\delta(\omega_{\mathbf{p}_1'} + \omega_{\mathbf{p}_2'} - \omega_{\mathbf{k}'} - \omega_{\mathbf{p}_1}) \delta_{\mathbf{p}_2 \mathbf{k}} \right. \\
& \times \left(\hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{p}_2'} - \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_1' \mathbf{k}'} \delta_{\mathbf{q}_2' \mathbf{p}_1} \right. \\
& + \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}_2'} + \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}_1'} \\
& - \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_2' \mathbf{k}'} - \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{k}'} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_1' \mathbf{p}_1} \Big) \\
& - \delta(\omega_{\mathbf{k}} + \omega_{\mathbf{p}_1'} - \omega_{\mathbf{p}_2} - \omega_{\mathbf{p}_1}) \delta_{\mathbf{p}_2' \mathbf{k}'} \left(\hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{k}} \right. \\
& - \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_1' \mathbf{p}_2} \delta_{\mathbf{q}_2' \mathbf{p}_1} + \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \delta_{\mathbf{q}_2 \mathbf{k}} \\
& + \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \delta_{\mathbf{q}_2 \mathbf{p}_1'} - \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_2' \mathbf{p}_2} \\
& \left. \left. - \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_1' \mathbf{p}_1} \right) \right]. \tag{A.5}
\end{aligned}$$

Due to symmetries of the sum one can rewrite this equation as

$$\begin{aligned}
\frac{d}{dt}(\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}'}) &= \frac{i}{\hbar} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_1', \mathbf{p}_2'} U(\mathbf{p}_1 - \mathbf{p}_2) \left(\hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{p}_2 \mathbf{k}} - \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{p}_2} \hat{a}_{\mathbf{p}_1} \delta_{\mathbf{p}_1' \mathbf{k}'} \right) \\
& - \frac{2\pi}{\hbar^2} \sum_{\substack{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_1', \mathbf{p}_2' \\ \mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_1', \mathbf{q}_2'}} U(\mathbf{p}_1 - \mathbf{p}_2) U(\mathbf{q}_1 - \mathbf{q}_2) \left[\delta(\omega_{\mathbf{p}_1'} + \omega_{\mathbf{p}_2'} - \omega_{\mathbf{k}'} - \omega_{\mathbf{p}_1}) \right. \\
& \times \delta_{\mathbf{p}_2 \mathbf{k}} \left(\hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{p}_2'} - \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_1' \mathbf{k}'} \delta_{\mathbf{q}_2' \mathbf{p}_1} \right. \\
& + 2\hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}_2'} - 2\hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_2' \mathbf{k}'} \Big) \\
& - \delta(\omega_{\mathbf{k}} + \omega_{\mathbf{p}_1'} - \omega_{\mathbf{p}_2} - \omega_{\mathbf{p}_1}) \delta_{\mathbf{p}_2' \mathbf{k}'} \left(\hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{k}} \right. \\
& - \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_1' \mathbf{p}_2} \delta_{\mathbf{q}_2' \mathbf{p}_1} + 2\hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{q}_2}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \delta_{\mathbf{q}_2 \mathbf{k}} \\
& \left. \left. - 2\hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_1}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \delta_{\mathbf{q}_2' \mathbf{p}_2} \right) \right]. \tag{A.6}
\end{aligned}$$

Next we take the expectation value from both sides of the evolution equation. Using Wick's theorem [3] on the second-order terms we have

$$\begin{aligned}
 & \langle \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \rangle \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{p}_2'} - \langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \rangle \delta_{\mathbf{q}_1' \mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}_1} \\
 &= \left(\langle \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \langle \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{k}'} \rangle + \eta \langle \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{k}'} \rangle \langle \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \right) \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{p}_2'} \\
 &\quad - \left(\langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_2} \rangle \langle \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_1} \rangle + \eta \langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_1} \rangle \langle \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \rangle \right) \delta_{\mathbf{q}_1' \mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}_1} \\
 &= \langle \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \langle \hat{a}_{\mathbf{k}'}^\dagger \hat{a}_{\mathbf{k}'} \rangle \left(\delta_{\mathbf{q}_1' \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{k}'} + \eta \delta_{\mathbf{q}_1' \mathbf{k}'} \delta_{\mathbf{q}_2' \mathbf{p}_1} \right) \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{p}_2'} \\
 &\quad - \langle \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{p}_1'} \rangle \langle \hat{a}_{\mathbf{p}_2}^\dagger \hat{a}_{\mathbf{p}_2'} \rangle \left(\delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{p}_2'} + \eta \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{p}_2'} \right) \delta_{\mathbf{q}_1' \mathbf{k}'} \delta_{\mathbf{q}_2 \mathbf{p}_1}, \\
 & \langle \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \rangle \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{k}} - \langle \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \rangle \delta_{\mathbf{q}_1' \mathbf{p}_2} \delta_{\mathbf{q}_2' \mathbf{p}_1} \\
 &= \langle \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \langle \hat{a}_{\mathbf{p}_2}^\dagger \hat{a}_{\mathbf{p}_2} \rangle \left(\delta_{\mathbf{q}_1' \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{p}_2} + \eta \delta_{\mathbf{q}_1' \mathbf{p}_2} \delta_{\mathbf{q}_2' \mathbf{p}_1} \right) \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{k}} \\
 &\quad - \langle \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{p}_1'} \rangle \langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle \left(\delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{k}} + \eta \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{k}} \right) \delta_{\mathbf{q}_1' \mathbf{p}_2} \delta_{\mathbf{q}_2' \mathbf{p}_1}.
 \end{aligned}$$

Similarly

$$\begin{aligned}
 & \langle \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{k}'} \rangle \delta_{\mathbf{q}_2 \mathbf{p}_2'} - \langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \rangle \delta_{\mathbf{q}_2' \mathbf{k}'} \\
 &= \left(\langle \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \langle \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{k}'} \rangle \langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_1} \rangle + \eta \langle \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{k}'} \rangle \langle \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_1} \rangle \right) \delta_{\mathbf{q}_2 \mathbf{p}_2'} \\
 &\quad - \left(\langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_2} \rangle \langle \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_1} \rangle \langle \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{p}_1} \rangle + \eta \langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_1} \rangle \langle \hat{a}_{\mathbf{p}_2'}^\dagger \hat{a}_{\mathbf{q}_2} \rangle \langle \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \right) \delta_{\mathbf{q}_2' \mathbf{k}'} \\
 &= \langle \hat{a}_{\mathbf{k}'}^\dagger \hat{a}_{\mathbf{k}'} \rangle \langle \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_1'} \rangle \left(\delta_{\mathbf{q}_1' \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{k}'} + \eta \delta_{\mathbf{q}_1' \mathbf{k}'} \delta_{\mathbf{q}_2' \mathbf{p}_1} \right) \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{p}_2'} \\
 &\quad - \langle \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_1'} \rangle \langle \hat{a}_{\mathbf{p}_2}^\dagger \hat{a}_{\mathbf{p}_2'} \rangle \left(\delta_{\mathbf{q}_1 \mathbf{p}_2} \delta_{\mathbf{q}_2 \mathbf{p}_1'} + \eta \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{p}_2'} \right) \delta_{\mathbf{q}_1' \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{k}'}, \\
 & \langle \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{q}_2'}^\dagger \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{q}_1} \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{p}_2} \rangle \delta_{\mathbf{q}_2 \mathbf{k}} - \langle \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{q}_1'}^\dagger \hat{a}_{\mathbf{p}_1} \hat{a}_{\mathbf{q}_2} \hat{a}_{\mathbf{q}_1} \rangle \delta_{\mathbf{q}_2' \mathbf{p}_2} \\
 &= \langle \hat{a}_{\mathbf{p}_2}^\dagger \hat{a}_{\mathbf{p}_2} \rangle \langle \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_1'} \rangle \left(\delta_{\mathbf{q}_1' \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{p}_2} + \eta \delta_{\mathbf{q}_1' \mathbf{p}_2} \delta_{\mathbf{q}_2' \mathbf{p}_1} \right) \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{k}} \\
 &\quad - \langle \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{p}_1} \rangle \langle \hat{a}_{\mathbf{p}_1'}^\dagger \hat{a}_{\mathbf{p}_1'} \rangle \langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} \rangle \left(\delta_{\mathbf{q}_1 \mathbf{k}} \delta_{\mathbf{q}_2 \mathbf{p}_1'} + \eta \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{k}} \right) \delta_{\mathbf{q}_1' \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{p}_2}.
 \end{aligned}$$

Taking this into account the evolution equation becomes

$$\begin{aligned}
 \frac{d}{dt} \langle \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}'} \rangle &= \frac{2i}{\hbar} \sum_{\mathbf{p}_1} U(\mathbf{p}_1 - \mathbf{k}) n_{\mathbf{p}_1} (n_{\mathbf{k}'} - n_{\mathbf{k}}) - \frac{4\pi}{\hbar^2} \sum_{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_1', \mathbf{p}_2'} \sum_{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_1', \mathbf{q}_2'} \\
 &\quad \times U(\mathbf{p}_1 - \mathbf{p}_2) U(\mathbf{q}_1 - \mathbf{q}_2) \left[\delta(\omega_{\mathbf{p}_1'} + \omega_{\mathbf{p}_2'} - \omega_{\mathbf{k}'} - \omega_{\mathbf{p}_1}) \delta_{\mathbf{p}_2 \mathbf{k}} \right. \\
 &\quad \times \left(\frac{1}{2} n_{\mathbf{k}'} n_{\mathbf{p}_1} (\delta_{\mathbf{q}_1' \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{k}'} + \eta \delta_{\mathbf{q}_1' \mathbf{k}'} \delta_{\mathbf{q}_2' \mathbf{p}_1}) \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{p}_2'} \right. \\
 &\quad \left. - \frac{1}{2} n_{\mathbf{p}_1} n_{\mathbf{p}_2'} (\delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{p}_2'} + \eta \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{p}_2'}) \delta_{\mathbf{q}_1' \mathbf{k}'} \delta_{\mathbf{q}_2' \mathbf{p}_1} \right. \\
 &\quad \left. + n_{\mathbf{k}'} n_{\mathbf{p}_1} n_{\mathbf{p}_1'} (\delta_{\mathbf{q}_1' \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{k}'} + \eta \delta_{\mathbf{q}_1' \mathbf{k}'} \delta_{\mathbf{q}_2' \mathbf{p}_1}) \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{p}_2'} \right]
 \end{aligned}$$

$$\begin{aligned}
& - n_{\mathbf{p}_1} n_{\mathbf{p}_1'} n_{\mathbf{p}_2'} \left(\delta_{\mathbf{q}_1 \mathbf{p}_2'} \delta_{\mathbf{q}_2 \mathbf{p}_1'} + \eta \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{p}_2'} \right) \delta_{\mathbf{q}_1 \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{k}'} \\
& - \delta \left(\omega_{\mathbf{k}} + \omega_{\mathbf{p}_1'} - \omega_{\mathbf{p}_2} - \omega_{\mathbf{p}_1} \right) \delta_{\mathbf{p}_2' \mathbf{k}'} \\
& \times \left(\frac{1}{2} n_{\mathbf{p}_1} n_{\mathbf{p}_2} \left(\delta_{\mathbf{q}_1 \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{p}_2} + \eta \delta_{\mathbf{q}_1' \mathbf{p}_2} \delta_{\mathbf{q}_2' \mathbf{p}_1} \right) \delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{k}} \right. \\
& - \frac{1}{2} n_{\mathbf{p}_1'} n_{\mathbf{k}} \left(\delta_{\mathbf{q}_2 \mathbf{p}_1'} \delta_{\mathbf{q}_1 \mathbf{k}} + \eta \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{k}} \right) \delta_{\mathbf{q}_1' \mathbf{p}_2} \delta_{\mathbf{q}_2' \mathbf{p}_1} \\
& + n_{\mathbf{p}_2} n_{\mathbf{p}_1} n_{\mathbf{p}_1'} \left(\delta_{\mathbf{q}_1' \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{p}_2} + \eta \delta_{\mathbf{q}_1' \mathbf{p}_2} \delta_{\mathbf{q}_2' \mathbf{p}_1} \right) \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{k}} \\
& \left. - n_{\mathbf{p}_1} n_{\mathbf{p}_1'} n_{\mathbf{k}} \left(\delta_{\mathbf{q}_1 \mathbf{k}} \delta_{\mathbf{q}_2 \mathbf{p}_1'} + \eta \delta_{\mathbf{q}_1 \mathbf{p}_1'} \delta_{\mathbf{q}_2 \mathbf{k}} \right) \delta_{\mathbf{q}_1' \mathbf{p}_1} \delta_{\mathbf{q}_2' \mathbf{p}_2} \right) \Big],
\end{aligned} \tag{A.7}$$

where we introduced notations $n_{\mathbf{p}} = \langle \hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{p}} \rangle$.

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